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Ex-10

Explore Antivirus Detection Techniques

Aim:

To study and understand different techniques used by antivirus software to detect malicious programs and protect systems from threats.

Procedure:

- Install any antivirus software (e.g., Windows Defender or others).
- Update the antivirus database to the latest version.
- Scan a sample file or folder using the antivirus.
- Observe how the antivirus detects threats (signature-based, heuristic, behavior-based).
- Analyze the scan report and identified threats.

Task-1:

Task-2:

McAfee Associates, Inc. started the first AV software implementation in 1987. It was called "VirusScan," and its main goal at that time was to remove a virus named "Brain" that infected John McAfee's computer. Later, other companies joined in the battle against viruses. AV software was called scanners, and they were command-line software that searched for malicious patterns in files.

Since then, things have changed. AV software nowadays uses a Graphical User Interface (GUI) to perform scans for malicious files and other tasks. Malware programs have also expanded in scope and now target victims on Windows and other operating systems. Modern AV software supports most devices and platforms, including Windows, Linux, macOS, Android, and iOS. Modern AV software has improved and become more intelligent and sophisticated, as they pack a bundle of versatile features, including Antivirus, Anti-Exploit, Firewall, Encryption tool, etc.

We will be discussing some AV features in the next task.

Answer the questions below

What does AV mean?

Antivirus — ✓ Correct Answer

Which PC Antivirus vendor implemented the first AV software on the market?

McAfee — ✓ Correct Answer

Antivirus software is a _____-based security solution.

Host — ✓ Correct Answer

Task-3:

Other common features

The following are some common features found in AV products:

- A self-protection driver to guard against malware attacking the actual AV.
- Firewall and network inspection functionality.
- Command-line and graphical interface tools.
- A daemon or service.
- A management console.

Answer the questions below

Which AV feature analyzes malware in a safe and isolated environment?

Emulator — ✓ Correct Answer

An _____ feature is a process of restoring or decrypting the compressed executable files to the original.

unpacker — ✓ Correct Answer

Read the above to proceed to the next task, where we discuss the AV detection techniques.

No answer needed — ✓ Correct Answer

Task-4:

100%

Task 1 Introduction

Welcome to Intro to AV

You can also connect via RDP, make sure you deploy the AttackBox or connect to the VPN.

Use the following credentials below.

Machine IP: `MACHINE_IP` Username: `thm` Password: `TryHackM3`

Connect to the VM via the RDP client

```
user@machine$ xfreerdp /v:MACHINE_IP /u:thm /p:TryHackM3
```

Answer the questions below

Once you've deployed the VM, it will take a few minutes to boot up. Then, progress to the next task!

No answer needed ✓ Correct Answer

Task-5:

The output shows we improved our Yara rule to reduce the false-positive results. That was a simple example of how AV software works. Thus, AV software vendors work hard to fight against malware and improve their products and database to enhance the performance and accuracy of results.

The drawback of the signature-based detection is that files will have a different hash value if the binary is modified. Therefore, it is easy for someone to bypass signature-based detection techniques if they know what AV software looks for and how to analyze binaries, as shown in later rooms.

Answer the questions below

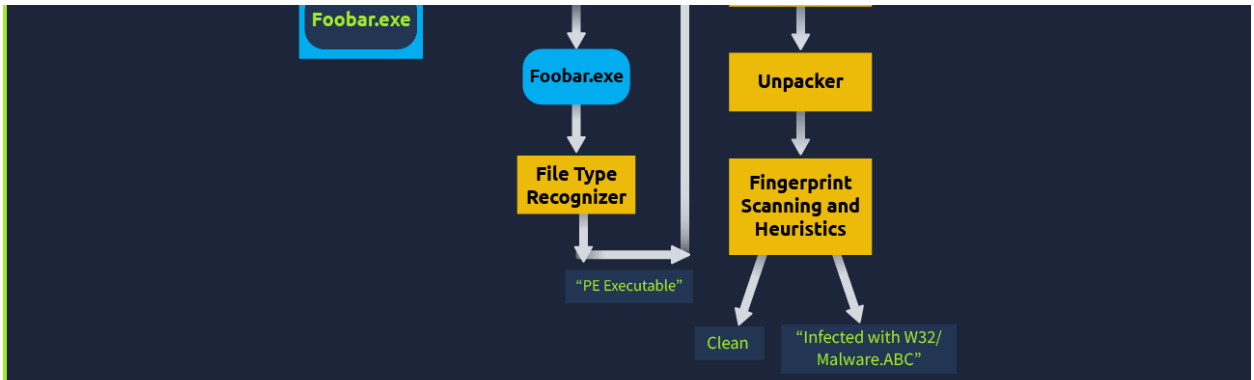
What is the `sigtool` tool output to generate an MD5 of the `AV-Check.exe` binary?

f4a974b0cf25dca7fbce8701b7ab3a88:6144:AV-Check.exe ✓ Correct Answer

Use the strings tool to list all human-readable strings of the AV-Check binary. What is the flag?

THM{Y0uC4nC-5tr16s} ✓ Correct Answer

Task-6:



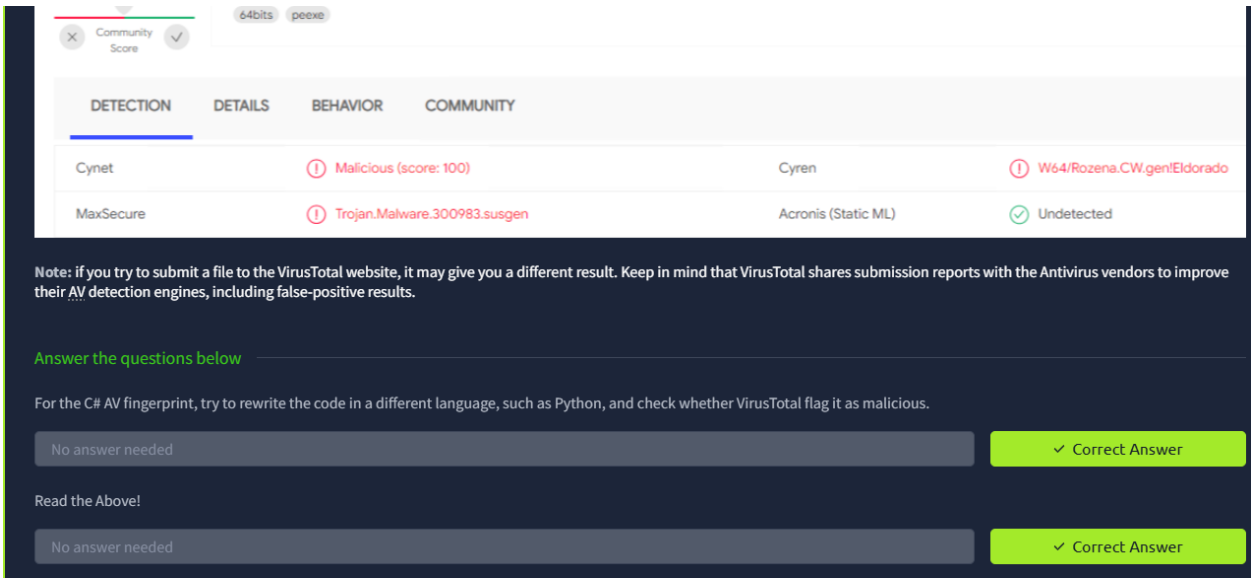
In the diagram, you can see a suspicious `Foobar.zip` file is passed to AV software to scan. AV software recognizes that it is a compressed file (.zip). Since the software supports .zip files, it applies an un-archiver feature to extract the files (`Foobar.exe`). Next, it identifies the file type to know which module to work with and then performs a PE parsing operation to pull the binary's information and other characteristic features. Next, it checks whether the file is packed; if it is, it unpacks the code. Finally, it passes the collected information and the binary to the AV engine, where it tries to detect if it is malicious and gives us the result.

Answer the questions below

Which detection method is used to analyze malicious software inside virtual environments?

Dynamic Detection ✓ Correct Answer

Task-7:



Note: if you try to submit a file to the VirusTotal website, it may give you a different result. Keep in mind that VirusTotal shares submission reports with the Antivirus vendors to improve their AV detection engines, including false-positive results.

Answer the questions below

For the C# AV fingerprint, try to rewrite the code in a different language, such as Python, and check whether VirusTotal flag it as malicious.

No answer needed ✓ Correct Answer

Read the Above!

No answer needed ✓ Correct Answer

Task-8:

Task 8 Conclusion

Recap

In this room, we covered Antivirus software and its detection approaches. As a red teamer, it is important to know how AV software works and detects malicious applications to be able to implement bypass techniques.

We also discussed the static detection technique in detail and showcased how ClamAV, an open-source Antivirus, detects malicious files using static analysis. Additionally, we showed how to create your own database and use Yara rules to detect malicious files that are not detected by the official database.

Once we obtain access to a target, it is essential to enumerate the target machine before performing further actions, such as privilege escalation or lateral movement. The reason is to not trigger alerts for suspicious activities, which may cause losing access to the target machine. Thus, we introduced two methods to practice AV fingerprinting using public and private tools.

Answer the questions below

Congrats on completing the room, and keep learning!

No answer needed

✓ Correct Answer

Result:

Antivirus detection techniques effectively identify both known and unknown threats using multiple methods, ensuring better system security and protection from malware.